

CLOUD COMPUTING

Instead of buying and maintaining your own computers/servers, you rent them from someone else over the internet.

1. What it actually is It's using someone else's computers (servers) to store data, run software, or process information — all accessed through the internet.

2. The 3 main types you'll hear about

- **IaaS** (Infrastructure as a Service) — You rent raw computers/storage. *Example: AWS EC2*
- **PaaS** (Platform as a Service) — You rent a ready-made environment to build apps. *Example: Google App Engine*
- **SaaS** (Software as a Service) — You just use the software directly. *Example: Gmail, Netflix, Zoom*

MICROSOFT AZURE CLOUD

Azure is Microsoft's giant cloud platform — basically a massive collection of services you can rent over the internet to build, run, and manage almost anything digital.

What it actually is Azure is Microsoft's cloud computing platform — it has **200+ services** running in Microsoft's data centers worldwide. You access all of it through the internet.

2. The main things Azure offers

- **Virtual Machines** — Rent a computer in the cloud (no physical hardware needed)
- **Storage** — Store files, images, videos safely online
- **Web Hosting** — Host your website or app
- **Databases** — Store and manage your app's data
- **AI & Machine Learning** — Build smart apps without being an AI expert
- **Security** — Microsoft handles protection and compliance
- **Analytics** — Analyze huge amounts of data easily

Azure is actually **all three** — IaaS, PaaS, and SaaS combined in one platform.

IaaS (Infrastructure as a Service) Yes, Azure *has* IaaS — like **Virtual Machines**, where you rent raw computing power and manage it yourself. *You control the OS, software, everything.*

PaaS (Platform as a Service) Azure also offers PaaS — like **Azure App Service**, where you just upload your code and Azure handles the servers, OS, and runtime for you. *You focus only on your app.*

SaaS (Software as a Service) Azure also delivers SaaS — like **Microsoft 365 (Office, Teams, Outlook)** which runs on Azure infrastructure. *You just use the software, no management needed.*

Advantages of Cloud Computing

1. **Cost Saving** — Pay only for what you use, no hardware investment
2. **Scalability** — Instantly scale up or down based on your needs
3. **Accessibility** — Access your data from anywhere with internet
4. **Security** — Enterprise-grade protection managed by experts
5. **Auto Backup** — Your data is always backed up and never lost
6. **Auto Updates** — Always get the latest version without manual effort
7. **Eco-Friendly** — Shared resources mean less energy waste
8. **Collaboration** — Multiple people can work together in real time
9. **Fast Deployment** — Launch servers or apps in minutes, not weeks
10. **Reliability** — 99.9% uptime with no single point of failure

Real-Time Use Cases of Cloud Computing

1. **E-Commerce (Amazon / Flipkart)** During Big Billion Sale, millions shop at once — cloud instantly scales up servers to handle the traffic and scales back down after the sale.
2. **Video Streaming (Netflix / YouTube)** Every video you watch is stored and delivered from cloud servers — no cloud means no streaming, anywhere in the world.
3. **Healthcare (Hospitals & Health Apps)** Patient records, scans, and reports are stored securely on cloud — doctors can access them from anywhere anytime in emergencies.
4. **Banking & UPI Payments (GPay / PhonePe)** Every time you send money, cloud processes millions of transactions per second securely and instantly without failure.
5. **Online Gaming (PUBG / Xbox Cloud Gaming)** Games run on powerful cloud servers — even a basic phone can play high-end games because the heavy processing happens in the cloud, not on your device.

